Assessment: Linux File Permissions Quiz

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📝 Part 1: Practical Scenarios

Q1: Access Control Reset

Scenario: A file currently has `-rwxr-xr-x` permissions. You need to modify it so the **Owner** has Read/Write access, the **Group** has Read-only access, and **Others** have no access at all.

- **Command:** `chmod 640 <filename>`
- **Logic:** Owner (4+2=6), Group (4), Others (0).

Q2: Script Execution

Scenario: A script is set to `644`. You want to make it executable for all users while maintaining the existing Read/Write permissions for the owner.

- **Command:** `chmod 755 <filename>`
- **Logic:** Owner (6+1=7), Group (4+1=5), Others (4+1=5).

Q3: Collaborative Logging

Scenario: A log file must be writable by both the **Owner** and the **Group**, but it should remain readable by everyone on the system.

- **Command:** `chmod 664 <filename>`
- **Logic:** Owner (4+2=6), Group (4+2=6), Others (4).

📝 Part 2: Binary & Octal Conversion

Q4: Binary String to Octal

Question: If a file's binary permission string is `110 100 000`, what is the corresponding `chmod` numeric command?

- **Answer:** `chmod 640`
- **Logic:** `110` (4+2=6), `100` (4), `000` (0).

Q5: The "Zero" Permission

Question: What occurs if you execute `chmod 000 secret.txt`? Can the file owner still read the file using `cat` without using `sudo`?

- **Answer:** No.
- **Logic:** `000` removes all permissions for all users, including the owner. Attempting to `cat` the file will result in a "Permission denied" error. Only a superuser (`sudo`) can bypass this.

📝 Part 3: Symbolic & Numeric Mapping

Q6: File Initialization

Task: Create a file named `Audit.txt` and assign it `rwxr-x---` permissions.

- **Command:** `touch Audit.txt && chmod 750 Audit.txt`
- **Logic:** Owner (rwx=7), Group (r-x=5), Others (—=0).

Q7: Numeric to Symbolic (741)

Question: What is the symbolic notation for the octal value `741`?

- **Answer:** `rwxr----x`
- **Logic:** 7 (rwx), 4 (r–), 1 (–x).

Q8: Numeric to Symbolic (500)

Question: What is the symbolic notation for the octal value `500`?

- **Answer:** `r-x------`
- **Logic:** 5 (r-x), 0 (—), 0 (—).

Q9: Symbolic to Numeric (-r-xrw-r–)

Question: What is the numeric equivalent of the symbolic string `-r-xrw-r--`?

- **Answer:** `564`
- **Logic:** `r-x` (4+1=5), `rw-` (4+2=6), `r--` (4).

📊 Reference: The 4-2-1 Rule

Octal	Binary	Symbolic	Description
7	111	`rwx`	Full Access (Read, Write, Execute)
6	110	`rw-`	Read and Write
5	101	`r-x`	Read and Execute
4	100	`r--`	Read Only
2	010	`-w-`	Write Only
1	001	`--x`	Execute Only
0	000	`---`	No Permissions